Scalar–Tensor Hydrodynamics

A Modified Gravity Framework for Vacuum Viscosity
and Cyclic Vacuum Cosmology

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Revised Formalism — Version 3.6

Abstract

We propose a hydrodynamic extension of General Relativity in which the vacuum is modeled as a dynamic, high-energy scalar medium described by a field ϕ. Rather than treating the vacuum as a fixed geometric background plus a constant cosmological term, the framework assigns the vacuum its own stress-energy, pressure, and dissipative response. The formulation separates the Einstein geometric tensor from the vacuum viscous stress tensor, introduces a dynamical scalar-field sector, replaces a fixed cosmological constant with a field-dependent pressure contribution, and allows the effective gravitational coupling to depend on the local vacuum state. The framework is intended to investigate whether cosmological expansion, galactic dynamical anomalies, and the transition between particulate matter and vacuum-field excitations can arise from one underlying medium.

This document is a theoretical proposal, not an established physical theory. The equations below define a research program whose consistency, stability, conservation laws, and observational predictions must be demonstrated.

Preface: What This Paper Is For

The problem

Two of the largest numbers in cosmology are placeholders. Roughly a quarter of the universe's contents is labelled *dark matter*: a substance inferred entirely from its gravitational pull on things we can see, and never once detected directly. Most of the remainder is labelled *dark energy*, which is less a substance than a name for an observation — that cosmic expansion is speeding up when everything we understand says it should be slowing down. Together these two labels cover something like ninety-five per cent of what the universe is made of. Neither has an agreed physical identity.

Standard cosmology handles this competently. It adds an invisible cold matter component, and a constant term to Einstein's equations, and the resulting fit to observation is superb — good enough that the model deserves the confidence it has. But a superb fit obtained by inserting two unexplained quantities is not the same thing as an explanation. It is a description with two holes in it, and the holes have names.

This paper asks whether both holes might have one filling.

A note on where these numbers came from

There is a precedent in this subject for inserting a quantity because the answer is expected rather than because anything has been measured. In 1917 Einstein added a term to his field equations — the cosmological constant, Λ — for no reason beyond his conviction that the universe ought to be static, and the inconvenient fact that his equations refused to make it so. No observation supported the term. It was a number chosen to produce an anticipated result. When Hubble demonstrated a little over a decade later that the universe is expanding after all, Einstein abandoned it.

It is worth being precise about what happened next, because the common retelling is wrong in a way that matters. Λ returned in 1998, but not as Einstein's unmeasured assumption quietly surviving. It

returned because observations of distant supernovae showed cosmic expansion to be accelerating, and Λ was the simplest term capable of describing that. The acceleration is real and measured. What Λ does not do is explain it: no theory predicts the constant's value, and the various attempts to calculate it from first principles fail by a margin discussed below. The term records what the universe does and is silent on why.

Dark matter has an entirely separate and considerably stronger pedigree. Fritz Zwicky noticed in 1933 that galaxies in the Coma cluster move too fast for the visible mass to hold them together. Vera Rubin's rotation-curve measurements in the 1970s found the same discrepancy inside individual galaxies. Gravitational lensing, the structure of the cosmic microwave background, and the separation of mass from gas in colliding clusters have since confirmed it from independent directions. The evidence is not thin, and this document does not dispute any of it.

The objection is therefore not that these quantities were invented. It is that in both cases a measured discrepancy was given a name, and the name was then promoted to a substance. Λ is a parameter fitted to the expansion history rather than derived from anything. Dark matter is a material known solely by its gravity, which forty years of increasingly sensitive detectors have failed to find by any other means. Each is a placeholder that has been in service long enough to be mistaken for an answer.

If the framework set out here is even approximately correct, nothing observational changes. The rotation curves remain exactly as measured. So do the lensing maps, the microwave background, and the supernova data establishing acceleration. What dissolves is the pair of substances invoked to account for them: they stop being two unknown constituents of the universe and become two behaviours of one medium. The measurements survive intact. Only their cause is reassigned.

Where our two best theories disagree

General Relativity and quantum field theory are the most thoroughly confirmed descriptions of nature ever constructed, and they are known to be mutually incompatible. In ordinary practice this causes little trouble, because the two rarely describe the same situation: one governs the very large, the other the very small, and the regimes where both matter at once — the first instants of the universe, the interior of a black hole — are exactly the regimes we cannot observe.

There is one glaring exception, and it is directly relevant here. The two theories overlap on the question of what the vacuum is, and there they do not disagree by a little. General Relativity treats empty space as genuinely empty: a smooth, featureless backdrop with no properties of its own. Quantum field theory describes the same emptiness as the most active object in physics — fields in their lowest state, never still, producing effects that have been measured in the laboratory for seventy years.

Ask both theories what that activity should weigh, and the answers differ by something between sixty and a hundred and twenty orders of magnitude, depending on where the calculation is cut off. This is not a discrepancy to be tidied away with a correction factor. It is routinely described as the worst quantitative prediction in the history of science, and it has stood unresolved for decades.

The observation that motivates this document is that the sharpest disagreement between our two best theories is a disagreement about whether the vacuum is a thing. One says no. The other says emphatically yes, and has the laboratory results to support it. What follows takes the second answer seriously and asks a further question that neither theory addresses: if the vacuum is a thing, does it also have mechanical properties — density, pressure, flow, resistance to being stirred — and if so, what would we expect to see?

The idea

General Relativity treats spacetime as a stage. It can be curved and stretched by what sits on it, but it has no substance of its own: no density, no pressure, no internal motion. The vacuum, in that picture, is the absence of things.

This framework treats the vacuum as a *medium* instead. Not empty, but full: a continuous field, here called the Ocean, with its own energy, its own pressure, its own internal flow, and its own resistance to being stirred. Matter and light are not objects placed in a void; they are disturbances in something that was already there.

The appeal of this move is that media do things stages cannot. They flow. They drag. They carry waves. They can be stirred, and they push back when you stir them. Each of those behaviours is a candidate for a phenomenon currently attributed to something unknown:

- A galaxy rotating inside a medium that swirls along with it experiences stress from that swirl. Stress gravitates. The extra pull currently assigned to a halo of undetected particles might instead be the vacuum's own response to being rotated.
- A medium with internal pressure that can change character over time can drive expansion without a constant pasted into the equations by hand. What is now called dark energy would be the Ocean in a particular state rather than a number in a table.
- And — this is the part developed furthest here — a medium has a smallest meaningful size. Below some scale, it cannot be disturbed in a localized way at all. That gives a principled distinction between what counts as a particle and what counts as a ripple.

The ambition is that dark matter and dark energy stop being two mysteries and become two behaviours of one thing.

What this document is not

It is not a theory. It is a proposal with its homework listed.

The equations below are consistent in form, but consistency in form is cheap. A physical theory has to survive conservation laws, stability analysis, the recovery of tested results in the appropriate limits, and quantitative agreement with observations that were not used to build it. None of that has been done here. Section 11 is an itemized list of the work required, and several items on it could terminate the framework outright. They are stated plainly rather than buried, because a proposal that cannot specify how it would fail is not worth the paper it occupies.

How this document came to exist

The framework began as private speculation, written to a standard that required only that the pieces appear to fit. That is a much weaker standard than the one it has since been held to, and the gap between the two turned out to be where the interesting work was.

Three revisions have followed, and they share a shape.

The first corrected a structural error. In the original formulation the vacuum's viscous response *replaced* the Einstein tensor — the geometric heart of General Relativity. This cannot work: a mathematical identity guarantees that the Einstein tensor is conserved, and substituting something else for it breaks the conservation of energy and momentum. The fix was to keep the geometry intact and add the vacuum's stress alongside it rather than in its place.

The second removed a number. The original drew the line between matter and vacuum at 0.511 MeV — the rest energy of the electron. Above the line, particles; below it, vacuum. This was an acknowledged placeholder, and it had three problems: the value was imported rather than derived, it was observer-dependent (two observers moving relative to one another would disagree about which side of the line a given thing fell), and it was an abrupt cutoff where physics prefers continuity.

The replacement drops the number entirely and asks instead how small a disturbance the medium can hold. Every medium has such a scale — in a laboratory superfluid it is called the healing length, the shortest distance over which the medium can bend without tearing. Disturbances larger than it spread out and remain part of the medium. Disturbances smaller than it stay put, as lumps. The matter/vacuum distinction becomes a property of the Ocean rather than a line drawn across it.

The result worth noting is what happened next. Calibrating that scale to present conditions puts it at roughly 386 fm, which is exactly the size an electron occupies by the relevant measure. The electron, on this account, is simply the lightest thing small enough to be a lump in the Ocean — and everything lighter, photons and neutrinos included, is necessarily a ripple instead. The number that had been an

unexplained input came back out as a consequence. That is the kind of exchange that makes a framework worth continuing.

The third responded to news from outside. In September 2026 a proof was announced that a fluid with internal friction can, under smooth external stirring, tear itself into a singularity in finite time — a result about ordinary fluid dynamics with no connection to this framework whatsoever. It remains unverified and does not transfer directly to a relativistic medium. But it removed an assumption this framework had been resting on without noticing: that viscosity is enough to keep a continuum well-behaved forever.

The repair was already present. The smallest-scale limit introduced in the second revision is precisely what stops a disturbance concentrating without bound — but only if the medium resists compression rather than welcoming it. That condition had never been written down. It is now a stated requirement, and if the Ocean turns out to have the other sign, the framework fails as written.

Each revision has the same form: something chosen by hand is removed, and something forced by the structure takes its place. A substituted tensor, then an imported number, then an unexamined assumption. That a framework can absorb that process repeatedly is not evidence that it is true. It is evidence that it is the right kind of thing to keep testing — which is a lower bar, and the only one this document claims to clear.

A note on method, since it is unusual enough to be worth recording. The revisions above were worked out in dialogue with a large language model, used as an interlocutor rather than an author: a way of being argued with at the speed of thinking. There is a small irony in the third revision, where the result that forced the change was itself produced by a machine, formally verified by another, and is still waiting on human confirmation.

What would kill it

Three things, named here so that a reader does not have to hunt for them.

The scale gap. For the Ocean to explain galactic rotation it must be able to act across thousands of light years. For it to place the matter boundary at the electron it must have structure at the scale of a hundredth of an atom. These two requirements differ by about thirty-two orders of magnitude. They are set by different parts of the mathematics, so this is not an outright contradiction — but a framework that merely tolerates such a gap has not explained anything, and this is the most serious unresolved problem in the document.

A route out now exists, and it is recorded as a route rather than a result. A hurricane is hundreds of miles across, but that size is not a property of air. It belongs to the storm: to how much energy and spin were gathered in one place. Air itself has only small-scale properties. If the same holds here, the galactic scale is not built into the Ocean at all. A galaxy is not an object that stirs the medium; it is a region of the medium carrying excess energy and excess current, of which the visible stars are the dense middle and the wide slow flow is the outskirts. The Ocean would then need only one built-in size, the small one, together with the ability to carry very long, slow disturbances at almost no cost. Laboratory superfluids have that ability for a known reason, a symmetry of the underlying field. The price is that the field must be a richer object than the one written down below, and that the calculation showing such a disturbance reproduces observed galaxies has not been done. Until it is, the gap stays on this list.

The sign. As above: if the Ocean's self-interaction has the wrong sign, nothing arrests collapse and the framework is not viable in its present form. This is a single calculation away from being settled either way.

The preferred frame. Describing the vacuum as a medium gives it a state of rest, and a state of rest is something Einstein spent a career removing from physics. Real superfluids do exactly this, so it is not absurd — but the experimental limits on any such effect are extraordinarily tight, and the framework must show it stays inside them.

To which should be added a possibility that no calculation will resolve: the agreement between the Ocean's smallest scale and the size of the electron may simply be a coincidence. The Standard Model

already accounts for the electron's mass by other means. Whether these are two descriptions of one fact or two unrelated facts that happen to agree is, at present, unknown.

What follows is the technical statement of all of this. It is written for a reader comfortable with tensor notation, but the argument it makes is the one above: that the emptiness between things may not be empty, and that a great deal of what we currently call unknown may be the behaviour of a single medium that has been sitting in plain view the whole time.

1 The Field Equation

The governing equation for the "Ocean" metric is formulated as Einstein geometry coupled to a dynamic scalar vacuum medium. The central revision relative to the original formulation is that the viscous response is treated as a *stress contribution* rather than as a replacement for the geometric Einstein tensor:

$$G_{\mu\nu} + P(\phi)\, g_{\mu\nu} \;=\; \kappa(\phi)\Big[\, T^{\text{matter}}_{\mu\nu} + T^{\phi}_{\mu\nu} + \Pi_{\mu\nu} \,\Big], \qquad \kappa(\phi) = \frac{8\pi G(\phi)}{c(\phi)^4}. \tag{1}$$

Here $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu}$ remains the geometric Einstein tensor; $P(\phi)$ is the dynamic vacuum pressure; $T^{\phi}_{\mu\nu}$ is the stress-energy of the scalar field; and $\Pi_{\mu\nu}$ is its hydrodynamic dissipative stress.

2 Tensor Definitions

2.1 Geometric Einstein tensor

The geometric sector retains the standard covariant structure of General Relativity:

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}. \tag{2}$$

This separation is essential because the contracted Bianchi identity gives $\nabla^{\mu} G_{\mu\nu} = 0$. Any additional vacuum stresses must therefore satisfy the corresponding total conservation requirement (Section 9).

2.2 Scalar vacuum stress-energy $T^{\phi}_{\mu\nu}$

The vacuum is represented by a dynamical scalar field rather than a purely phenomenological density. Let

$$X \equiv -\tfrac{1}{2}\, \nabla_{\alpha}\phi\, \nabla^{\alpha}\phi, \tag{3}$$

and let $L(\phi, X)$ be the scalar-field Lagrangian. Its stress-energy tensor is

$$T^{\phi}_{\mu\nu} \;=\; L_{,X}\, \nabla_{\mu}\phi\, \nabla_{\nu}\phi \;+\; L\, g_{\mu\nu}. \tag{4}$$

This term gives the Ocean a genuine dynamical energy density and pressure. The specific choice of L determines whether the field behaves as a conventional scalar, a k-essence-like medium, or a more specialized superfluid effective theory. Section 7 depends on this choice directly.

A candidate refinement. The candidate route for the scale hierarchy (Section 7.7, item (5)) requires more structure than a single real scalar. Take the medium to be a complex field with a global $U(1)$ symmetry,

$$\Phi = \rho\, e^{i\theta}, \qquad \theta \;\to\; \theta + \text{const}, \tag{5}$$

and identify the ϕ of the preceding expressions with the phase θ. The shift symmetry forbids a potential for θ, so that $L(\phi, X) \to L(X)$ and the long-wavelength excitation is a gapless Goldstone phonon, as in a laboratory superfluid. The modulus ρ is the natural home of the self-interaction of Section 9.1, and its excitation is heavy.

This is a condition the route depends on, not an adopted change, and it has a cost that propagates. An exact shift symmetry forbids any dependence on the value of θ itself. The state-dependent quantities $P(\phi)$, $\kappa(\phi)$, $\xi(\phi)$, $\eta(\phi)$ and $\zeta(\phi)$ would have to be re-expressed as functions of the modulus ρ or of X, and the phase transitions of the Heartbeat cosmology would become transitions of ρ. The same applies to the quartic potential of Section 9.1, which becomes a potential for ρ, and to the mass-gap entry of the table in Section 7, which vanishes identically for θ. The route also brings the framework's field content closer to the complex-field description of Huang and collaborators [3] and to the phonon description of Berezhiani and Khoury [4]; the distinctive claim of Section 10, the partition with a retained fundamental geometry, is unaffected.

2.3 Vacuum viscosity tensor $\Pi_{\mu\nu}$

The viscous response represents the dissipative stress response of the scalar medium to local flow. For a relativistic effective fluid, a first-order constitutive form is

$$\Pi_{\mu\nu} = -2\,\eta(\phi)\,\sigma_{\mu\nu} \;-\; \zeta(\phi)\,\theta\,h_{\mu\nu}, \tag{6}$$

where $\eta(\phi)$ is the shear viscosity, $\zeta(\phi)$ the bulk viscosity, $\theta = \nabla_\alpha u^\alpha$ the expansion scalar, $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ the spatial projector orthogonal to the medium four-velocity u^μ, and $\sigma_{\mu\nu}$ the shear tensor.

In regions with coherent vorticity or differential flow, $\Pi_{\mu\nu}$ can generate additional gravitational stress. The model therefore investigates whether the vacuum response can reproduce effects conventionally assigned to dark-matter halos. The term *rotational support* is used here as a hypothesis about the gravitational effect of the vacuum stress response; viscosity itself is not assumed to behave as additional particulate mass.

2.4 Scalar pressure $P(\phi)$

$P(\phi)$ replaces a fixed cosmological constant as the phenomenological vacuum-pressure contribution. It is a function of the state of the underlying scalar field rather than a universal constant:

$$\Lambda g_{\mu\nu} \;\longrightarrow\; P(\phi)\,g_{\mu\nu}. \tag{7}$$

The *Heartbeat* hypothesis proposes that the scalar field can evolve through alternating dynamical phases. An effective positive-pressure / negative-pressure transition may then produce expansion ("diastole") and contraction ("systole") phases. A complete model must derive this behavior from a field potential or equation of state rather than prescribing an oscillating pressure by hand.

2.5 Effective coupling $\kappa(\phi)$

The coupling between the vacuum state and ordinary stress-energy is allowed to depend on the local scalar density:

$$\kappa(\phi) = \frac{8\pi G(\phi)}{c(\phi)^4}. \tag{8}$$

$G(\phi)$ represents a field-dependent effective gravitational coupling. The quantity $c(\phi)$ is treated initially as an effective propagation speed associated with the vacuum medium. If it is interpreted as a genuine variable causal speed, the theory must specify the resulting causal structure and recover the observed local value c_0 in the appropriate limit.

3 Scalar-Field Dynamics

The field ϕ must have an independent equation of motion. A generic covariant form is

$$\Box\phi \;=\; \frac{\partial V_{\text{eff}}}{\partial\phi} \;+\; C(\phi, T, \nabla\phi, \ldots), \tag{9}$$

where $V_{\text{eff}}(\phi)$ is an effective potential and C represents permitted couplings to matter, curvature, or hydrodynamic variables. The potential and coupling structure determine the vacuum phases, stability, characteristic sound speed, and possible cyclic behavior.

4 The Ocean Interpretation

The framework interprets the observable universe as a dynamical region of a deeper vacuum medium. The Ocean is not required to begin at the Big Bang. Instead, the Big Bang can be interpreted as a state transition, a phase transition, or a rapidly expanding solution of the underlying field equations.

The observable universe is therefore not treated as the ontological boundary of reality. It is the portion of the deeper continuous vacuum field accessible to observation. This distinction is a conceptual premise of the framework and must ultimately be connected to a mathematically well-defined global solution.

5 Cosmological Heartbeat

The proposed *Heartbeat* cosmology associates cosmic expansion and contraction with the evolution of ϕ. A schematic cycle is

ϕ-state transition $\rightarrow$ pressure reversal $\rightarrow$ expansion $\rightarrow$ field evolution $\rightarrow$ contraction $\rightarrow$ new phase.

The model does not require that each cycle create the underlying vacuum field. Instead, each cosmological epoch may be a dynamical state of the same deeper medium. Whether such a cycle can avoid singularities, entropy accumulation, and instabilities is an open mathematical question.

6 Galactic Dynamics

In the galactic regime, the hypothesis is that spatial gradients, vorticity, and coherent flow of the vacuum field generate an additional effective stress. The observable circular velocity would then be determined by the combined baryonic and vacuum-field gravitational response:

$$v_c^2(r) \;=\; v_b^2(r) \;+\; v_{\phi,\mathrm{eff}}^2(r). \tag{10}$$

Here v_b is the baryonic contribution and $v_{\phi,\mathrm{eff}}$ denotes the effective contribution produced by the scalar field and its stress response. The goal is not to insert an arbitrary halo profile, but to derive $v_{\phi,\mathrm{eff}}(r)$ from ϕ, $\Pi_{\mu\nu}$, and the field equations.

One structure, not two. The decomposition above is bookkeeping and should not be read as a statement that a galaxy acts on the medium from outside. In this framework a galaxy is a region of the Ocean carrying excess localized energy and elevated current. Its baryonic content is the $\mathscr{Q} \gg 1$ population of that region and the extended flow is the $\mathscr{Q} \ll 1$ population, in the sense of Section 7; the two are the dense interior and the diffuse exterior of a single solution of the field equations. This differs from treatments in which galaxies are external sources perturbing the superfluid [3].

Two consequences follow. First, the kiloparsec scale is a property of the individual solution, fixed by the energy and angular momentum concentrated in it, and need not be a parameter of the medium; this is the basis of the candidate route of Section 7.7, item (5). Second, if the solution is fixed by its baryonic content — the question posed by the Size task of Section 11 — the baryonic distribution determines $v_{\phi,\mathrm{eff}}(r)$ with no freedom adjusted galaxy by galaxy. The observed radial acceleration relation has that form: across rotationally supported galaxies the total acceleration is fixed by the baryonic acceleration alone, with little or no intrinsic scatter and no sign of a second parameter [9, 10]. That is a target to be derived, not a result.

It also sharpens the exposure recorded in Section 10. A contribution fixed by the baryons is precisely the kind that risks being parallel to the baryonic gravitational field, and a parallel contribution is excluded by Milky Way vertical kinematics [7, 8]. The single-structure reading therefore needs a vacuum contribution whose magnitude is set by the baryons while its direction is not slaved to theirs. Whether the field equations permit that is not known.

7 The Localization Threshold

7.1 Why a fixed threshold fails

The earlier formulation partitioned energy between sectors by a declared inequality:

$$E > 0.511\,\text{MeV} \;\Rightarrow\; \text{particulate matter sector}, \qquad E \le 0.511\,\text{MeV} \;\Rightarrow\; \text{scalar-fluid density } \phi.$$

This has three defects, only the first of which is usually noticed.

(1) **It is an unexplained parameter.** The value is imported from the electron rest mass, which the framework does not derive.

(2) **It is not covariant.** E is frame-dependent. A boost changes which sector a given excitation belongs to, so the partition is observer-dependent — unacceptable in a theory built on the covariant field equation (1).

(3) **It is discontinuous.** A step function requires an exchange term coupling $T^{\text{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$. Combined with the variable coupling $\kappa(\phi)$, this makes the conservation conditions of Section 9 essentially impossible to satisfy.

The revision removes all three by replacing the declared threshold with a property of the vacuum field's own excitation spectrum.

7.2 The excitation spectrum of ϕ

Perturb the field about a background configuration,

$$\phi(x) = \bar{\phi}(x) + \delta\phi(x), \tag{11}$$

and let u^μ be the four-velocity of the vacuum medium with spatial projector $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$. Expanding the Lagrangian $L(\phi, X)$ of Section 2.2 to second order in $\delta\phi$ gives a quadratic action of the form

$$S_2 = \frac{1}{2}\int \sqrt{-g}\left[\mathscr{K}\left(u^\mu \partial_\mu \delta\phi\right)^2 - L_{,X}\, h^{\mu\nu}\partial_\mu\delta\phi\,\partial_\nu\delta\phi - V''_{\text{eff}}\,\delta\phi^2 - \mathscr{G}\left(h^{\mu\nu}\partial_\mu\partial_\nu\delta\phi\right)^2\right], \tag{12}$$

with $\mathscr{K} \equiv L_{,X} + 2XL_{,XX}$. The final term is a gradient-stiffness (quantum-pressure) operator. In a superfluid effective theory it arises from the same expansion; here it must be *derived* from the chosen L or from an underlying superfluid EFT, not assumed (see Section 7.7).

Varying (12) and evaluating in the medium rest frame yields the dispersion relation

$$\boxed{\omega^2(k,\phi) \;=\; \Omega^2_{\text{gap}}(\phi) \;+\; c_s^2(\phi)\,k^2 \;+\; \frac{\hbar^2}{4m_\star^2(\phi)}\,k^4} \tag{13}$$

7.3 The three characteristic scales

Every coefficient in (13) is fixed by the Lagrangian rather than chosen:

Quantity	Definition	Physical meaning
Sound speed	$c_s^2 = L_{,X}/\mathscr{K}$	Propagation speed of coherent Ocean disturbances
Mass gap	$\Omega^2_{\text{gap}} = V''_{\text{eff}}(\bar{\phi})/\mathscr{K}$	Curvature of the effective potential at the local vacuum state
Effective inertia	$m_\star = m_\star(\mathscr{G}, \mathscr{K})$	Stiffness of the field against short-wavelength bending

From these follows the single scale that carries the whole section — the *healing length*:

$$\xi(\phi) \;=\; \frac{\hbar}{m_\star(\phi)\,c_s(\phi)}. \tag{14}$$

ξ is the shortest distance over which the vacuum field can vary without the gradient term dominating. It is the Ocean's grain size.

7.4 The localization criterion

Define the dimensionless quantity

$$\mathscr{Q}(\phi,k) \;\equiv\; \xi^2(\phi)\, h^{\mu\nu} k_\mu k_\nu. \tag{15}$$

Because $h^{\mu\nu}$ projects onto the rest frame of the medium, $\mathscr{Q}$ is constructed covariantly from the excitation's four-momentum and the medium's own state. It replaces the frame-dependent E of the old formulation.

$$\mathscr{Q} \ll 1 \implies \omega \approx c_s k \qquad \text{phononic branch; delocalized; contributes to } T^{\phi}_{\mu\nu}, \tag{16}$$

$$\mathscr{Q} \gg 1 \implies \omega \approx \hbar k^2/2m_\star \qquad \text{particle branch; localized; contributes to } T^{\text{matter}}_{\mu\nu}. \tag{17}$$

The crossover sits at $k_c = 1/\xi(\phi)$, and the corresponding energy is

$$E_c(\phi) = \hbar\,\omega(k_c) \;\approx\; \alpha\, m_\star(\phi)\, c_s^2(\phi), \qquad \alpha = \mathscr{O}(1). \tag{18}$$

The partition is smooth, not switched. Define a monotonic weighting $f(\mathscr{Q})$ with $f(0) = 0$ and $f(\infty) = 1$ — for example $f = \mathscr{Q}/(1+\mathscr{Q})$ — and split the mode integral:

$$T^{\text{matter}}_{\mu\nu} = \int d^3k\, f(\mathscr{Q})\, t_{\mu\nu}(k), \qquad T^{\phi}_{\mu\nu} = \int d^3k\, \big[1 - f(\mathscr{Q})\big]\, t_{\mu\nu}(k). \tag{19}$$

Since both sectors are populations of the *same* field, their sum is conserved identically. No exchange term is required, and the consistency conditions of Section 9 are no longer obstructed by the threshold.

7.5 Recovering 0.511 MeV as a calibration

Requiring $E_c(\phi_0) \approx 0.511\,\text{MeV}$ in the present vacuum state fixes

$$\xi(\phi_0) = \frac{\hbar c}{E_c} \approx 386\,\text{fm}, \tag{20}$$

which is the reduced Compton wavelength of the electron. This is a derived consequence rather than a coincidence. A particle of mass m has Compton wavelength $\bar{\lambda}_C = \hbar/mc$; it can localize within the medium only if $\bar{\lambda}_C \lesssim \xi_0$, which requires

$$m \;\gtrsim\; \frac{\hbar}{c\,\xi_0} \;=\; m_e. \tag{21}$$

The electron is therefore the lightest excitation the present vacuum can hold as a localized object. Anything lighter has a Compton wavelength longer than the Ocean's grain and cannot be a lump in it — it is necessarily a coherent field excitation. This is offered as the framework's explanation for why photons and neutrinos behave as field phenomena while the electron is the floor of the particulate sector. The value $0.511\,\text{MeV}$ is demoted from axiom to measurement: it is the readout of $\xi(\phi_0)$, not an input.

7.6 Consequences

The threshold is local and dynamical. $E_c = E_c(\phi(x))$, so regions in different vacuum states have different matter thresholds. This connects the present section to the Heartbeat cosmology: a phase transition of ϕ is also a transition in what the vacuum is *able to hold* as matter. Where ξ grows, the localization floor rises and previously particulate states are no longer supported.

It predicts a varying electron mass. If ξ evolves, so does the effective m_e. This is a genuine observational handle — and a genuine constraint.

It introduces a preferred frame. The dispersion relation (13) is stated in the rest frame of u^μ. This is honest — real superfluids do exactly this — but it is a departure from exact local Lorentz invariance and must be reconciled with precision tests.

7.7 Open items specific to this section

(1) **Derive the gradient-stiffness term** $\mathscr{G}$ from a covariant action rather than importing it from the non-relativistic Bogoliubov form. Without this, $m_\star$ and hence ξ are not defined by the theory.

(2) **Stability.** Require $L_{,X} > 0$ and $\mathscr{K} = L_{,X} + 2XL_{,XX} > 0$; otherwise $c_s^2 < 0$ (gradient instability) or the kinetic term flips sign (ghost).

(3) **Lorentz-violation bounds.** Show $c_s \to c_0$ in the local limit to the precision demanded by clock-comparison and photon/neutrino time-of-flight tests, and demonstrate that the k^4 term is suppressed at accessible energies.

(4) **Varying-m_e bounds.** Quasar absorption spectra constrain drift in m_p/m_e at roughly the 10^{-5}–10^{-6} level over cosmological time; the Oklo natural reactor constrains it locally. Any Heartbeat evolution of ξ must be negligibly slow in the present epoch and confined to transition epochs.

(5) **The scale hierarchy.** Galactic-scale effects require the scalar's Compton range to be of order a kiloparsec, implying $\Omega_{\rm gap} \sim 10^{-26}\,\mathrm{eV}$. But $\xi_0 \approx 386\,\mathrm{fm}$ implies a stiffness scale near $0.5\,\mathrm{MeV}$. These differ by roughly thirty-two orders of magnitude. They are set by different operators — $V''_{\rm eff}$ versus $\mathscr{G}$ — so this is not immediately inconsistent, but a credible model must *explain* the separation rather than merely tolerate it. This is the most serious open problem in the revised section.

Candidate route. The estimate $\Omega_{\rm gap} \sim 10^{-26}\,\mathrm{eV}$ assumes that the kiloparsec scale is intrinsic to the medium. On the single-structure reading of Section 6 it is instead a property of the individual galactic solution. The requirement on the medium then weakens from a tuned value to an upper bound,

$$\Omega_{\rm gap} \lesssim 10^{-26}\,\mathrm{eV}, \tag{22}$$

which $\Omega_{\rm gap} = 0$ satisfies. A vanishing gap is protected, rather than tuned, if ϕ is the phase of a complex field with a $U(1)$ shift symmetry (5); a small nonzero gap from soft breaking of that symmetry would be technically natural. The medium would then carry one intrinsic length scale, ξ_0, and the hierarchy would be a ratio between a parameter of the medium and the size of a particular solution, which is not a naturalness problem. In the same construction the heavy modulus is the expected origin of the k^4 term in (13), which would bear on item (1).

None of this is established. The route trades the hierarchy for three obligations: to show that the symmetry survives the couplings of equation (1); to say what fixes the size of a galactic solution; and to pass the vertical-acceleration test in the form stated in Section 6. It also carries the cost recorded in Section 2.2. Until those are met the hierarchy stands as an open problem.

(6) **Relation to the Standard Model.** The Higgs mechanism already accounts for m_e via a Yukawa coupling. The framework must state whether ξ_0 and the electron Yukawa are independent quantities that happen to agree, or whether one is meant to determine the other.

8 Limiting Regimes

Regime	Required behavior	Physical objective
GR / Solar System	$\phi \to \phi_0$; $G(\phi) \to G_0$; $c(\phi) \to c_0$; vacuum stresses negligible	Recover tested General Relativity
Galactic	$\Pi_{\mu\nu}$ and scalar gradients become significant	Explain rotation curves without particulate dark matter
Cosmological	$P(\phi)$ and $T^{\phi}_{\mu\nu}$ contribute on large scales	Generate expansion history and effective dark-energy behavior
Transition / Heartbeat	ϕ evolves between dynamically distinct states; $\xi(\phi)$ shifts	Provide a mechanism for expansion/contraction cycles and for changes in the localization floor
High-energy	Mode occupation crosses $\mathscr{Q} \sim 1$	Test the derived localization threshold of Section 7

9 Conservation and Consistency Conditions

Because $\nabla^{\mu} G_{\mu\nu} = 0$, the complete right-hand side of (1) together with the pressure sector must satisfy a compatible conservation law. With variable $\kappa(\phi)$, the simple condition $\nabla^{\mu} T_{\mu\nu} = 0$ is generally not sufficient by itself. The scalar equation (9) and the matter coupling must be chosen so that the full theory is diffeomorphism-consistent.

The revised Section 7 assists here rather than obstructing: because $T^{\text{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$ are now weighted populations of a single field rather than two sectors joined by a step function, no discontinuous exchange term is introduced at the threshold.

9.1 Global regularity and the defocusing condition

A separate requirement, implicit in the previous version and made explicit here, is that the medium remain smooth for all time. The framework cannot appeal to dissipation alone to secure this. Viscosity is conventionally expected to drain an energy cascade before it concentrates without bound, but that expectation is not a theorem: finite-time blowup constructions for the forced three-dimensional Navier–Stokes equations were announced in September 2026 and, while they remain subject to independent verification and do not transfer directly to a relativistic scalar medium, they remove the intuition that a viscous continuum is automatically globally regular. The shear and bulk viscosities $\eta(\phi)$ and $\zeta(\phi)$ of Section 2.3 therefore cannot be assumed to guarantee what they were implicitly relied upon to guarantee.

The mechanism available to this framework is not dissipation but the ultraviolet cutoff supplied by Section 7. The gradient-stiffness term in the dispersion relation (13) grows as k^4 and therefore dominates the cascade at $k \sim 1/\xi(\phi)$, arresting concentration at the healing length. This is the mechanism by which a laboratory superfluid remains regular where the corresponding classical fluid description does not.

That arrest is conditional on the sign of the self-interaction. Writing the relevant quartic term of the effective potential as

$$V_{\text{eff}}(\phi) \supset \frac{\lambda}{4}\left(\phi^2 - v^2\right)^2, \tag{23}$$

the medium must be *defocusing*, $\lambda > 0$: compression must raise the energy and be resisted. The defocusing case is energy-coercive and the cutoff at ξ holds. In the focusing case $\lambda < 0$ the self-interaction rewards further compression, the cutoff does not arrest the cascade, and concentration accelerates instead. The sign is therefore a stability requirement of the same standing as $L_{,X} > 0$ and $\mathscr{K} > 0$, not a modelling preference.

A successful formulation should explicitly establish:

(1) covariance;

(2) total energy-momentum conservation;

(3) a well-posed scalar-field equation;

(4) absence of pathological ghosts or gradient instabilities;

(5) a stable low-density / weak-field limit;

(6) compatibility with precision tests of gravity and light propagation;

(7) a defocusing self-interaction, $\lambda > 0$, so that the healing-length cutoff of Section 7 arrests rather than accelerates concentration.

10 Relation to Existing Work

The proposal that the vacuum is a superfluid, and that dark matter and dark energy are two behaviours of it rather than two substances, is not new. This section states what is already established, so that the framework's actual claim can be isolated from the parts of it that are inherited.

10.1 Two families of superfluid-vacuum programme

Existing work divides on whether spacetime geometry is fundamental.

Emergent gravity. Volovik's programme [1] treats the quantum liquid as prior to spacetime: the metric, Lorentz invariance and gauge structure appear as low-energy collective behaviour of the condensate. Zloshchastiev's logarithmic superfluid vacuum theory [2] derives an induced gravitational potential from the vacuum wavefunction and concludes that dark matter and dark energy are manifestations of the same object and therefore cannot be independent. That conclusion is the one reached here. The mechanism is not: Zloshchastiev distinguishes his approach explicitly from models that work by modifying the stress-energy tensor of the Einstein equations.

Modified source. Huang and collaborators [3] treat a complex vacuum scalar field as a cosmic superfluid within standard General Relativity, identifying its energy density with dark energy and departures from vacuum density with dark matter. That work also treats galaxies as external sources perturbing the superfluid, and obtains the galactic halo together with the creation of vortices by galactic rotation — the closest existing precedent for the galactic sector of Section 6. Berezhiani and Khoury [4] construct an effective field theory in which dark matter is a superfluid with a galaxy-scale coherence length, described by phonons rather than collisionless particles, recovering MOND phenomenology on galactic scales. That model has since been disfavoured by Milky Way data [8]: it over-predicts vertical accelerations of disk stars even while reproducing the rotation curve. Viscous cosmology [5] models the dark sector as a single fluid whose bulk viscosity produces a negative effective pressure and thereby mimics dark energy. The logotropic models of Chavanis [6] belong to the same family, yielding a density profile that falls as r^{-1} and offering an alternative to MOND that accounts for the Tully–Fisher relation.

The present framework belongs to the second family. Equation (1) retains $G_{\mu\nu}$ as the geometric backbone and places the vacuum medium entirely on the right-hand side.

10.2 What is inherited and what is not

Inherited: the identification of the vacuum as a scalar condensate; the proposal that the dark sector is one medium rather than two substances; the use of bulk and shear viscosity to generate an effective pressure; the appearance of a coherence length in the galactic regime.

Inherited also, and more directly than the preceding list suggests: the generation of galactic structure by vorticity in a vacuum superfluid. Huang and collaborators [3] obtain vortices from galactic rotation within a nonlinear Klein–Gordon description in which galaxies act as external sources. Section 6 reaches for the same phenomenon through different machinery — a dissipative stress tensor $\Pi_{\mu\nu}$ entering the field equation (1) rather than a source term in a wave equation. Whether these are distinct physical mechanisms or one mechanism in two formalisms is not settled here, and a credible version of this framework must say which.

The premise that ordinary matter is itself an excitation of the vacuum medium is also not new, and it is worth being precise about who holds it. Huang [3] and Berezhiani and Khoury [4] are two-species: the condensate constitutes the dark sector while baryonic matter is a separate species coupled to it. Superfluid vacuum theory is not. Its stated paradigm is a background quantum liquid of which elementary particles are excitations — one substance, as here.

What remains distinctive is narrower, and is the conjunction of three things rather than any one of them. Superfluid vacuum theory obtains a single substance by making geometry emergent from the condensate. This framework attempts a single substance while retaining $G_{\mu\nu}$ as fundamental (Section 10 above), and the mechanism by which it does so is the covariant criterion of Section 7: the dimensionless quantity $\mathscr{Q} = \xi^2 h^{\mu\nu} k_\mu k_\nu$ of (15), with a smooth weighting $f(\mathscr{Q})$ splitting the mode integral between $T^{\text{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$. The consequence developed in Section 9 — that no discontinuous exchange term between the two is required, because both are populations of one field — follows from that construction. It removes an obstruction to the conservation conditions; it does not by itself establish them, which remains outstanding.

10.3 The electron-scale coincidence

The numerical agreement of Section 7 has been approached from the opposite direction. Working from a logotropic equation of state, Chavanis [6] recalls an expression for the cosmological constant in terms of the electron mass and the fundamental constants,

$$\Lambda = \frac{G^2 m_e^6}{\alpha^6 \hbar^4} \approx 1.36 \times 10^{-52}\,\mathrm{m}^{-2}, \tag{24}$$

a relation of Eddington type. There the electron mass fixes Λ; here the electron mass is the readout of $\xi(\phi_0)$. Both routes tie the electron scale to a vacuum-scale quantity, from opposite ends.

Two cautions attach to this. The relation is *recalled* rather than derived in that work, so the primary source lies earlier in the logotropic literature. And independent arrival at a coincidence is weak evidence that it is not accidental; it is not this framework's discovery, and the open question of Section 7.7, item (6), remains untouched by it.

10.4 Known deficiencies relative to current practice

Two points where this document is behind the literature it should be engaging.

(1) **First-order dissipative hydrodynamics.** The constitutive form of Section 2.3 is of Eckart–Landau type, which is known to be acausal and generically unstable in relativistic settings [11]. Current viscous cosmology uses causal second-order formulations of Israel–Stewart type [12]. A viable version of this framework must do the same.

(2) **Silence on vertical dynamics.** Section 6 states only a radial relation. Milky Way kinematics constrain the two components separately, and they do not want the same thing: the baryonic distribution supplies too little gravity to account for the rotation curve but enough to account for the vertical velocity dispersions near the Sun, so substantial enhancement is required radially and essentially none vertically [7, 8]. Superfluid dark matter fails this test because its phonon-mediated force is approximately parallel to the baryonic gravitational field and therefore enhances both components by the same factor [8]. Cold dark matter passes because its vertical acceleration is

dominated by baryons, with the halo subdominant. The resulting criterion applies to any model: the enhancement of vertical acceleration must be suppressed relative to the radial enhancement. Whether this framework is exposed depends on whether $v_{\phi,\mathrm{eff}}$ tracks the baryonic acceleration field — in which case it inherits the parallelism problem — or constitutes an independent source. Section 6 does not say which.

(3) **The status of $c(\phi)$.** A variable effective propagation speed is natural in the emergent-gravity family, where the metric is acoustic. It sits awkwardly alongside a fundamental $G_{\mu\nu}$. The framework must either justify $c(\phi)$ as a refractive effect within fixed geometry, or acknowledge that taking it as a genuine causal speed moves the proposal toward the emergent family and incurs that family's obligations.

10.5 The structural question

If the vacuum is a medium possessing density, pressure and a rest frame, it is reasonable to ask why it requires a background spacetime to occupy. The emergent-gravity family answers that it does not, and that geometry is the medium's own collective behaviour. This framework currently posits a medium and a geometry as distinct structures coupled through $\kappa(\phi)$. That is a coherent position, but it is a position, and it has not been argued for here.

11 Research Program and Testable Predictions

- Derive the field equations from a covariant action rather than introducing stress terms phenomenologically.
- Derive the effective equation of state $P(\phi)$ and determine whether a stable oscillatory cosmological solution exists.
- Solve the weak-field, stationary, rotating-galaxy limit and compare predicted rotation curves with baryonic distributions.
- Determine whether the same parameter set fits multiple galaxies without an independently chosen halo profile.
- Calculate the weak-field and post-Newtonian limits to verify recovery of Solar-System observations.
- Determine the propagation speed of scalar, gravitational, and electromagnetic perturbations, and identify whether $c(\phi)$ is a true causal speed or an effective refractive velocity.
- Test cosmological expansion, structure growth, gravitational lensing, and background-radiation constraints.
- **Derive the gradient-stiffness coefficient** $\mathscr{G}$, and with it the healing length $\xi(\phi)$ of (14), from the scalar Lagrangian.
- **Confront the predicted drift in** ξ — and hence in the effective electron mass — with quasar absorption spectra and Oklo constraints.
- **Resolve or explain the hierarchy** between the mass gap Ω_{gap} required by galactic dynamics and the stiffness scale implied by $\xi_0 \approx 386\,\mathrm{fm}$. The candidate route of Section 7.7, item (5), divides this into three tasks.
  - *Protection.* Reformulate the scalar sector with ϕ as the phase of a complex field (5), re-express P, κ, ξ, η, ζ and the quartic potential of Section 9.1 in terms of the modulus or of X, and determine whether the shift symmetry survives the matter coupling and the dissipative sector. If it does not, compute the gap that results.

  - *Size.* Determine what fixes the extent of a galactic solution, given that the medium supplies no kiloparsec scale of its own: which conserved quantities of the disturbance set it, and where their values originate.
  - *Rotation curves without per-galaxy freedom.* Derive $v_{\phi,\mathrm{eff}}(r)$ from the baryonic distribution alone, with no per-galaxy freedom, and compare with the observed radial acceleration relation [9, 10], including its universal acceleration scale $g_\dagger \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, which the framework must then account for. This must be done jointly with the vertical-acceleration item below, since a contribution fixed by the baryons is the kind most exposed to it.

- **Determine the sign of λ** from the chosen $L(\phi, X)$ and verify the defocusing condition of Section 9.1. A framework whose potential turns out to be focusing has no mechanism arresting concentration and is not viable as written.
- **Distinguish the galactic mechanism from its precedent.** Determine whether the rotational support of Section 6, generated by $\Pi_{\mu\nu}$, is physically distinct from the vortex formation obtained by Huang and collaborators [3] from a nonlinear Klein–Gordon equation with galaxies as external sources, or whether the two are the same mechanism expressed in different formalisms.
- **Establish the direction of the vacuum contribution.** Determine whether the acceleration produced by $\Pi_{\mu\nu}$ is parallel to the baryonic gravitational field, as a scalar-mediated MOND-like force is, or whether its anisotropic shear components yield a contribution with an independent orientation. This is the question on which the galactic sector turns: a parallel contribution enhances radial and vertical accelerations by a common factor and is excluded by local Milky Way data [7, 8], whereas an independent, quasi-spherical contribution is not. Then compute the predicted vertical acceleration explicitly and confront it, together with the rotation curve of Section 6, with stellar kinematics.
- **Reformulate the dissipative sector causally.** Replace the first-order constitutive relation of Section 2.3 with a second-order Israel–Stewart or BDNK form, and verify that the resulting system is causal and stable in the weak-field limit.
- **Settle the status of $c(\phi)$** as either a refractive effect within fixed geometry or a genuine causal speed, and state the consequences for the framework's relation to emergent-gravity formulations (Section 10).
- **Establish whether matter coupling constitutes a forcing term.** The known finite-time blowup constructions for viscous media require an external force injecting momentum; the unforced problem remains open. In this framework the candidate forcing is the matter sector acting on the vacuum through $\kappa(\phi)T^{\mathrm{matter}}_{\mu\nu}$ in (1). Determine whether that coupling can drive the medium to finite-time concentration, or whether the healing-length cutoff bounds it for any physically admissible matter distribution. This is the sharp form of the question raised in Section 5 as to whether a contraction phase terminates in a singularity or is arrested, and the answer turns on $\xi(\phi)$ and the sign of λ rather than on the viscosity coefficients.

12 Summary of the Revised Formalism

The revised Ocean framework is centered on a single idea: the vacuum is a physical, dynamical medium whose state can influence geometry, pressure, gravitational coupling, and hydrodynamic stress. The mathematical formulation keeps the Einstein tensor as the geometric backbone while adding a scalar-field stress sector, a viscous constitutive sector, a variable pressure contribution, and a field-dependent effective coupling.

The framework extends that idea to the matter sector itself. If the vacuum is a medium, it has a characteristic length over which it can heal; excitations shorter than that length are localized objects, and

excitations longer than it are motions of the medium. The distinction between matter and vacuum then becomes a statement about the medium's own excitation spectrum rather than an imposed energy cut.

The central hypothesis is that phenomena conventionally divided into dark energy, dark matter, and particulate matter may have a common origin in different dynamical regimes of one vacuum field. Establishing that claim requires deriving the relevant limits and demonstrating quantitative agreement with observations.

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Reference basis. Revised from an earlier formalism supplied by the author. That document defines the Ocean metric, the vacuum viscosity tensor, the scalar pressure term, the effective coupling $\kappa(\phi)$, the VSL interpretation, and the proposed 0.511 MeV matter threshold. The matter-threshold section of that document is superseded by Section 7 below.